MATH 174 - HW 1

Dani Nguyen

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1. (a) Forward substitution on $Ly = b$

$$\begin{aligned} y_1 &= 2 \\ -y_1 + y_2 &= -1 \\ 3y_1 - 3y_2 + y_3 &= 3 \end{aligned}$$

So, we have

$$\begin{aligned} y_1 &= 2 \\ y_2 &= -1 + x_1 \\ &= -1 + 2 \\ &= 1 \\ y_3 &= 3 - 3x_1 + 3x_2 \\ &= 3 - 3(2) + 3(1) \\ &= 0 \end{aligned}$$

(b) Backward substitution on $Ux = y$

$$\begin{aligned} -1x_1 + 2x_2 + x_3 &= 2 \\ 2x_2 - 2x_3 &= 1 \\ 5x_3 &= 0 \end{aligned}$$

So, we have

$$
\begin{aligned}
x_3 &= 0 \\
2x_2 &= 1 + 2x_3 \\
2x_2 &= 1 + 0 \\
x_2 &= \frac{1}{2} \\
-x_1 &= y_1 - 2x_2 - x_3 \\
x_1 &= -2 + 2(\frac{1}{2}) + 0 \\
x_1 &= -1
\end{aligned}
$$

(c) Compute $A = LU$

$$
A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix}
$$

$$
Ax = b : \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix} \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}
$$

The solution to $Ax = b$ is correct because

$$
\begin{aligned}
A\vec{x} &= \vec{b} \\
LU\vec{x} &= L\vec{y} \\
LU\vec{x} &= LU\vec{x}
\end{aligned}
$$

2. (a) The summation is from $j = 1$ to $i - 1$, there are $i - 1$ additions. There is also one more subtraction between b_i and the summation. So in total, there are i additions and subtractions.

   (b) There are $i - 1$ terms in the summation, therefore there are $i - 1$ multiplications.

3. (a) Gaussian elimination

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ -4 & 0 & 4 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ -4 & 0 & 4 & -1 \end{array}\right] R_2 = R_2 - \frac{1}{2}R_1$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & -6 & 6 & 1 \end{array}\right] R_3 = R_3 + 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & 0 & \frac{12}{5} & \frac{23}{5} \end{array}\right] R_3 = R_3 + \frac{12}{5}R_1$$

(b) Back substitution

$$\begin{aligned} \frac{12}{5}x_3 &= \frac{23}{12} \\ x_3 &= \frac{115}{144} \\ \frac{5}{2}x_2 &= \frac{3}{2}x_3 + \frac{3}{2} \\ x_2 &= \frac{3}{5}(\frac{115}{144}) + \frac{3}{5} \\ &= \frac{259}{240} \\ x_1 &= -1 + 4x_2 - 2x_3 \\ &= -1 + \frac{259}{60} - \frac{115}{72} \\ &= \frac{619}{360} \end{aligned}$$

4. (a) Gaussian elimination

$$\left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0.002 & -4 & -2 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & -4.01 & -2.01 \end{array}\right] R_2 = R_2 - 0.002R_1$$

Back substitution

$$\begin{aligned} -4.01x_2 &= -2.01 \\ x_2 &= \frac{201}{401} \text{ or } 0.501 \\ x_1 &= 3 - 4x_2 \\ &= \frac{399}{401} \text{ or } 0.995 \end{aligned}$$

(b) Gaussian elimination

$$\left[\begin{array}{cc|c} 0.002 & -4 & -2 \\ 1 & 4 & 3 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 0.002 & -4 & -2 \\ 0 & 2000 & 1000 \end{array}\right] R_2 = R_2 - 500R_1$$

Back substitution

$$\begin{aligned} 2000x_2 &= 1000 \\ x_2 &= \frac{1}{2} \\ 0.002x_1 &= -2 + 4x_2 \\ x_1 &= -1000 + 1000 \\ &= 0 \end{aligned}$$

(c) In a 3-digit rounding machine, $x = \frac{500}{501} = 0.998$ which is $\frac{\frac{500}{501} - 0.998}{\frac{500}{501}} = 4 \times 10^{-6}\%$ error.
For y, $y = \frac{1003}{2004} = .500$ which is $\frac{\frac{1003}{2004} - .5}{\frac{1003}{2004}} = 9.97 \times 10^{-4}\%$ error.
Therefore, $x = \frac{500}{501}$ is closer to the exact answer.

5. (a) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

(b) $\begin{bmatrix} -2 & 0 & 1 & -1 \\ -1 & 2 & 0 & 1 \\ 4 & -1 & -2 & -4 \\ 0 & 0 & 2 & 0 \end{bmatrix}$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 4 & -1 & -2 & -4 \\ 0 & 0 & 2 & 0 \end{bmatrix} R_2 = R_2 - \tfrac{1}{2}R_1$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & -1 & 0 & -6 \\ 0 & 0 & 2 & 0 \end{bmatrix} R_3 = R_3 + 2R_1$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -\frac{1}{4} & -\frac{21}{4} \\ 0 & 0 & 2 & 0 \end{bmatrix} R_3 = R_3 + \tfrac{1}{2}R_2$$

6.

7.

8.